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Advantages of continuous excavation in lightweight planetary robotic operations

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Abstract

Planetary excavator robots face unique and extreme engineering constraints relative to terrestrial counterparts. In space missions mass is always at a premium because it is the main driver behind launch costs. Lightweight operation, due to low mass and reduced gravity, hinders excavation and mobility by reducing the forces a robot can effect on its environment. This work shows that there is a quantifiable, non-dimensional threshold that distinguishes the regimes of lightweight and nominal excavation. This threshold is crossed at lower weights for continuous excavators (e.g. bucket-wheels) than discrete scrapers. This research introduces novel experimentation that for the first time subjects excavators to gravity offload (a cable pulls up on the robot with five-sixths its weight, to simulate lunar gravity) while they dig. A 300 kg excavator robot offloaded to 1/6 g successfully collects 0.5 kg/s using a bucket-wheel, with no discernible effect on mobility. For a discrete scraper of the same weight, production rapidly declines as rising excavation resistance stalls the robot. These experiments suggest caution in interpreting low-gravity performance predictions based solely on testing in Earth gravity. Experiments were conducted in GRC-1, a washed industrial silica-sand devoid of agglutinates and of the sub-75-micron basaltic fines that make up 40% of lunar regolith. The important dangers related to dust are thus not directly addressed. The achieved densities for experimentation are 1640 kg/m^3 (very loose/loose) and 1720 kg/m^3 (medium dense). This work develops a novel robotic bucket-wheel excavator, featuring unique direct transfer from bucket-wheel to dump bed as a solution to material transfer difficulties identified in past literature.

Keywords

Space robotics, planetary rovers, robotic excavation, reduced gravity, wheeled robots, traction.

1. Introduction

Excavation of regolith enables *in situ resource utilization* (ISRU) on the Moon and Mars. ISRU could potentially reduce the cost of exploration by producing consumables (including oxygen, water, and fuel) from native regolith as well as building earthwork infrastructure (such as trenches and berms) (Sanders and Larson, 2013).

In any space mission, mass is always at a premium because it is the main driver behind launch costs. Small excavators that can achieve mission goals are preferable to larger ones. Low mass machines operating in reduced gravity (one-sixth of Earth gravity on the Moon, one-third on Mars or Mercury) have limited weight available to produce traction or plunge tools into regolith. Traction and plunge force are limited to a fraction of the robot's weight. Engineering challenges associated with lightweight excavation necessitate a rethink of excavation configurations, possibly beyond the dozers, loaders, and excavators typical in terrestrial applications (Boles et al., 1993).

Excavation missions have started small. For example, the Surveyor, Viking, and Phoenix landers gathered samples of a few cubic centimeters at depths of a few centimeters with scoops mounted to relatively heavy landers. Future missions will likely escalate to excavating cubic meters' worth of regolith at depths of tens of centimeters using lightweight mobile robots (e.g. digging down to expose and collect water ice in polar regions of the Moon; Elphic et al., 2010). Eventually, excavation will scale up to production machines for ISRU. Having a configuration that scales well with increasing size and mass allows subsequent missions to re-use existing technology, learn from past difficulties, and reduce risk. This principle is exemplified in the similarities between Mars Sojourner, the subsequent Mars Exploration Rovers Opportunity and Spirit, and now the Mars Science Laboratory rover Curiosity. Robotic excavators with high productivity across a range of light weights are thus desirable. Another advantage of scalable small excavators is that

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deploying many smaller excavators rather than a few larger ones can also enhance operational redundancy.

The advantages of scalability and redundancy also bolster configurations that manage full load-haul-dump cycles with a single robot. Terrestrial mining often employs specialized machines for various stages of a load-haul-dump task; for example bucket-wheels for cutting and loading, and dump-trucks for hauling. Having multiple specialized vehicles for planetary excavation, though, requires multiplying the total mass devoted to baseline subsystems such as avionics, thermal, and so on, which is not mass-efficient at small scales (where these subsystems comprise a high percentage of the robot's total mass).

In addition to differences of lower system mass and reduced-gravity environments, planetary excavation also compels a different concept of operations than terrestrial excavation does. Planned planetary excavation operations, exemplified by the multiple cycles of field tests conducted jointly by the National Aeronautics and Space Administration (NASA) and the Canadian Space Agency (CSA), all consist of excavating horizontal strips of regolith from the terrain surface (Sanders and Larson, 2011; Boucher et al., 2011). This contrasts with the scope of terrestrial mining that also includes digging into the sides of piles during underground load-haul-dump operations as well as directly excavating the vertical or sloped walls of open pit mines; these operations involve more complex excavation bucket trajectories. To appreciate this reduction in scope in (at least early) planetary excavation, one can consider the nature of regolith, especially on the Moon. Vertical (i.e. 90°) slopes of lunar regolith are achievable to heights of up to 3 m, but the regolith's angle of repose in such a condition would be 45 (±3)° (Heiken et al., 1991). Digging into walls or piles could produce such vertical slopes, that would subsequently collapse to their angle of repose when disturbed by further excavation; in the vicinity of a small excavator that is already traction-limited, this risks burying the excavator in a mission-ending failure.

1.1. Planetary excavator configurations

Excavators can be classified as continuous or discrete, describing interactions with the soil while taking multiple cuts.

Continuous excavators stay continually in contact with the soil as they take multiple cuts. This necessitates having multiple cutting surfaces; by the time each surface or bucket has accumulated an appreciable amount of soil it clears the ground and the next, soil-free, bucket has already started cutting. Continuous excavators include bucket-wheels, bucket-chains, and elevating scrapers.

Discrete excavators are those that must break contact with the soil before starting a new cut; between cuts, the excavator may need to dump its load or clear the cutting surface, for example. These excavators fill one large bucket with a single cut. Discrete excavators include front-end loaders, backhoes, mining shovels, dozers, draglines, and open bowl scrapers. Discrete excavators can be further subdivided into **discrete scrapers** and **discrete scoopers**. Scrapers (including, by this definition, dozers and draglines) set a bucket depth and cutting angle and simply advance horizontally; the cutting edge has an ever-growing accumulation of soil as the bucket is filled. Scoopers are capable of more general cutting trajectories, penetrating into deeper soil and curling out of it.

A wide assortment of prototypes have been developed in recent years, of both the continuous and discrete *scraper* variety, specifically for planetary excavation and ISRU. This work compares the merits of these two classes of configuration, especially considering the effects of reduced gravity. Comparisons to discrete *scooping* are discussed as future work.

Discrete scrapers are very commonly proposed for planetary excavation. Example prototypes include NASA's Cratos (Caruso et al., 2008), an open bowl scraper with a central bucket between its tracks, as well as NASA's Chariot with LANCE bulldozer blade (King et al., 2010). Juno rovers (Theiss et al., 2010) have been equipped with a "load-haul-dump" bucket, though the fact that cut depth is the only controllable degree of freedom (DOF) makes it effectively a discrete scraper. NASA's Centaur 2 has been equipped with a full 2-DOF front-loader bucket, but experiments have exclusively involved horizontal cutting at set depths and cutting angles (Johnson et al., 2012). It is noteworthy that even configurations capable of discrete scooping have only been used as scrapers in the planetary excavation literature to date.

Among continuous planetary excavators, a past prototype bucket-wheel excavator for NASA was configured like a trencher with bucket-wheel and collection bin far apart and a conveyor in between (Muff et al., 2004). Such a configuration has had difficulty transferring regolith from bucket-wheel to collection bin, and as a result bucket-ladders have gained favor (Johnson and Van Susante, 2006). Winners of the NASA Regolith Excavation Challenge and subsequent Lunabotics mining competitions (competitions where lightweight excavators must collect as much regolith simulant as possible in a short period of time) have all employed bucket-ladders driven by exposed chains or belts (Mueller and Van Susante, 2011). A bucket-drum excavator, which is an adaptation of a bucket-wheel (Clark et al., 2009), has a novel regolith collection system with cutting buckets mounted directly around the outside of the collection drum. NASA's Regolith Advanced Surface Systems Operations Robot (RASSOR) has counter-rotating front and rear bucket drums, enabling it to balance horizontal excavation forces (Mueller et al., 2013).

The expansive design space of all these prototypes, and the wide variety of configurations it entails, makes it difficult to discern which particular features of any one-off

prototype make it either successful or not. For example, no prototype yet combines all the features of high productivity (including high payload capacity approaching 50% and relatively high speeds in the tens of centimeters per second) with low excavation resistance and lunar relevance. Prototypes have touched upon some of these ideas separately.

Several novel concepts and approaches to lunar and planetary excavation have been proposed that are complementary to the full system prototypes discussed above. Zacny and others (Craft et al., 2009; Szabo et al., 1998) advocate using percussive actuation of digging implements to reduce excavation resistance forces. Gustafson and Gertsch (2008) have studied the applicability of cutterhead wheels and forks for loosening frozen and compacted regolith in preparation for excavation. Bernold (1991) has suggested using small explosive charges to loosen compacted regolith. Zacny et al. (2008) have also proposed a pneumatic excavator that transports regolith particles using small quantities of low-pressure, high Mach number gas. Robots that excavate by filling up with regolith as they burrow into the ground have also been proposed (Kubota et al., 2007).

Trade studies have examined the applicability of various excavation robot options, specifically for lunar outpost site work. Boles et al. (1993) compared the probable required launch mass of several construction machine suites. They concluded that typical terrestrial excavation machines would not be as effective as tripod cranes, sweeper leveler/excavators, and other innovative vehicles. The work by Abu El Samid (2008) continued along the lines of Boles', concentrating on trade-offs between autonomy and teleoperation and between single-vehicle and team configurations. The author recommended a team of autonomous bulldozers for the task of berm building. The study by Mueller and King (2008) scored excavator designs on a number of quantitative and qualitative metrics and decided that a multi-purpose machine with a bulldozing blade and excavator arm was most appropriate for lunar site work. These trade studies begin with a suite of proposed excavator configurations (bulldozers vs front-end loaders, etc.) and compare their relative merit for lunar operations; in that sense, they espouse a top-down approach to configuration analysis. This work explores configurations for lightweight robotic excavators from the bottom up, starting with underlying variables (e.g. payload, excavation resistance force) that figure in an analytical framework of excavation tasks.

The parametric approach to exploring excavator configurations builds upon prior work by the authors. Sensitivity analyses were conducted on system parameters in both modeling and experimentation, and identified those that govern productivity (i.e. payload ratio, driving speed, etc.) (Skonieczny et al., 2014). Other parametric studies in the literature include modeling work by Zacny et al. (2010, 2013).

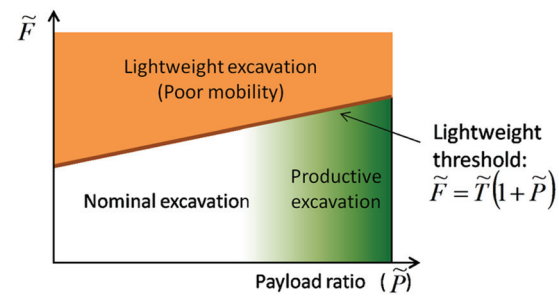

Fig. 1. The lightweight threshold distinguishes a regime in the operating space (governed by payload ratio and excavation forces) where an excavator is too light to produce enough traction to overcome resistance, resulting in degraded mobility. Payload ratio is a good predictor of excavation task productivity, so a key requirement is that an excavator must achieve high payload ratio without crossing the lightweight threshold. The lightweight threshold is developed quantitatively in Section 2.

1.2. Contribution, scope, and organization

The wide variability in configurations resulting from lunar excavation trade studies and prototype developments highlights the lack of consensus on appropriate configurations for productive lightweight excavators. This work develops a quantifiable, non-dimensional "lightweight threshold" that distinguishes lightweight from nominal excavation. It begins to constrain the design space by showing that the threshold is crossed at lower weights for continuous excavators than discrete scrapers, meaning continuous excavators are more appropriate for low-gravity operations.

Prior work by the authors has shown that payload ratio (the ratio of weight of regolith payload collected to empty robot weight) is a key parameter governing excavator productivity (Skonieczny et al., 2014). Thus, one of the central requirements of a planetary excavator is to achieve high payload ratio without crossing the lightweight threshold. Qualitatively, this can be depicted (Figure 1) as operating in the regime of productive excavation as opposed to the regime of lightweight excavation. A key contribution of this work is to show that discrete scraping, the most commonly proposed technique for planetary excavation, is not viable because it will stall the excavator before enough payload can be collected to be productive.

The scope of this work deals with excavators that produce thrust for cutting by developing traction. Excavation in gravity between one and one-sixth that of Earth is considered, to cover a range that includes Earth, Mars, Mercury, and the Moon. Digging on asteroids is outside the scope of this work. The scope of this work covers a range of weights that is relevant across multiple space mission scenarios; the range of robot mass considered spans 30 kg to 300 kg. Larger, more massive, machines are considered unlikely to satisfy mass budgets of near-future excavation missions. As machines get smaller than 30 kg or so, baseline components

that do not scale well (like computing and communications) take up an ever larger proportion of the mass, leaving little room for productive excavation tooling.

This paper develops a framework for analyzing excavator robots based on the definition of a lightweight number, L, in Section 2. Section 3 experimentally characterizes L for continuous and discrete scraper excavators as a basis for comparing these two classes of excavators in general. Section 4 demonstrates novel experimental techniques that are applied to directly showing the advantages of continuous excavation in lightweight situations. Section 5 presents a prototype continuous excavator that addresses some of the practical difficulties already discussed in this introduction. Conclusions and recommendations for future work are presented in Section 6.

2. A non-dimensionalized lightweight threshold

This work presents an analysis of the challenges associated with lightweight excavation. Because entrapment in regolith can be mission-ending, as it was for example for the Mars Exploration Rover Spirit, planetary excavation should nominally avoid impeding mobility. An excavator is thus considered too lightweight for its intended operations if excavation resistance forces exceed the traction that a robot is able to produce without unsafe slip and sinkage. This lightweight threshold, describing crossover into a regime of unsafe poor mobility, is set at 20% slip, as will be explained in Section 2.2.

As discussed in the introduction, a vital metric for excavators is productivity. Payload ratio governs excavator productivity (Skonieczny et al., 2014), so a key planetary excavator requirement is to achieve high payload ratio without crossing the lightweight threshold. The purpose of this section is to define the boundaries of the regimes introduced in Figure 1 quantitatively. Specifically, the relationships between payload accumulation, traction, and excavation resistance forces will be discussed. To preserve generality at various scales, non-dimensionalized quantities (denoted with tildes) are introduced for each of these terms.

2.1. Payload ratio

Payload ratio is the ratio of weight of regolith payload collected to empty robot weight; it is a measure of pound-for-pound regolith moving capacity that governs the productivity of lightweight robotic excavators. Terrestrial loaders and scrapers attain payload ratios as high as 80% to 100% (Caterpillar, 2002). Space systems are subject to additional constraints that make it challenging to attain payload values that high; the payload ratios of NASA's excavator prototypes top out at 30% (Caruso et al., 2008) to 40% (Mueller et al., 2013). A payload ratio of 50% is thus considered relatively high in this context. In this article, the non-dimensional quantity payload ratio is denoted $\tilde{P} = W_{payload}/W_{robot}$, where $W_{payload}$ and W_{robot} represent the weight of the currently collected regolith payload and of the empty robot, respectively.

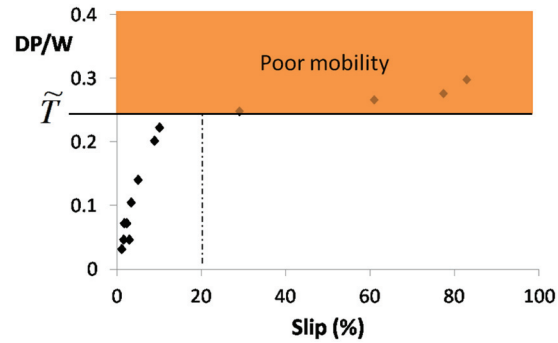

Fig. 2. Drawbar pull plateaus at around 20% slip and a vehicle enters a regime of poor mobility thereafter. This data, courtesy of NASA Glenn Research Center, characterizes the performance of a spring tire. Images of the tires themselves can be seen in Figure 9. $\tilde{T}$ is the value of DP/W at 20% slip.

2.2. Excavation thrust

Excavation thrust is the force supplied by an excavator that is available for cutting soil. In this work, excavation thrust is assumed to be provided by traction, as excavator configurations typically considered for space applications cut by driving forward.

A vehicle's drawbar pull, DP, is the net traction available for doing work, and is dependent upon slip. Drawbar pull at 20% slip, denoted P_{20}, is a good measure of tractive performance, as pull begins to plateau at around 20% slip for many wheels (or tracks) while negative effects such as sinkage increase (Turnage and Banks, 1989). An example of drawbar pull versus slip is shown in Figure 2 for a particular type of wheel called a spring tire; images of the tires themselves can be seen in Figure 9.

A non-dimensional quantity, P_{20}/W (drawbar pull at 20% slip, normalized by total weight), has been used as a benchmark metric for lunar wheel performance from the times of Apollo (Freitag et al., 1970) to today (Wong and Asnani, 2008; Wettergreen et al., 2010). In this work this ratio is called the **excavation thrust coefficient** and is denoted $\tilde{T} = P_{20}/W$.

The ratio P_{20}/W for any given wheel is approximately constant with changing load (i.e. W changing but scale and gravity constant), up to a critical loading (Freitag et al., 1970). Freitag demonstrated this by showing that P_{20} rises linearly with increasing W within a nominal range of W. It is expected that excavator wheels are designed properly to ensure an excavator with a full regolith payload still loads the wheels within this nominal range, so that $\tilde{T}$ remains constant throughout a load-haul-dump excavation task.

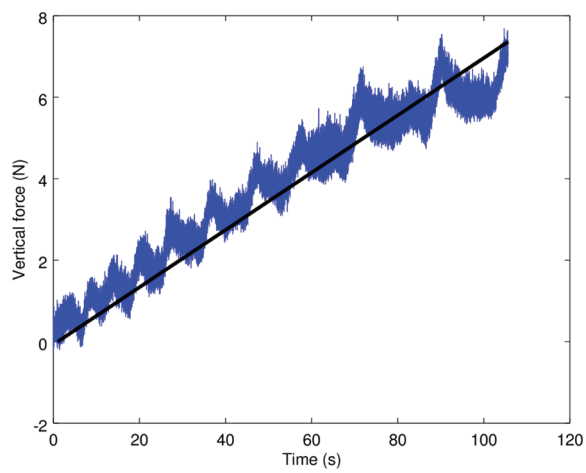

Fig. 3. $W_{payload}$ (black line) provides a good estimate of the vertical excavation forces (blue data) experienced during discrete scraping. Example data from "1.3 scale FP" test listed in Table 1.

2.3. Excavation resistance

Excavation resistance is the longitudinal force imparted on an excavator by cutting and collecting soil. Only the forces during the cut (once the excavation blade is already in the ground) are treated explicitly. Penetration forces are neglected but, as Blouin assumes these forces are of the same nature as the cutting forces (Blouin et al., 2001), they can be treated as analogous to cutting forces and subsumed by them. Crucially, resistance introduced by soil accumulation at a bucket's cutting edge (which increases the force required to move additional soil into the bucket) is accounted for as part of excavation resistance.

Forces imparted on an excavator include soil weight, soil–bucket interface friction and adhesion, as well as soil cutting resistance, and these forces in general may act along vectors with both vertical and longitudinal components (Hemami et al., 1994). In this work, excavation resistance is the sum of the longitudinal components of all these force contributions; this provides a direct counter to the longitudinal excavation thrust provided by traction. The vertical components of excavation forces cannot be ignored, as they alter the vertical loading on the excavator wheels and thus affect traction. In this work these vertical forces are indeed accounted for, by considering $W_{payload}$. The changing value of $W_{payload}$ is computed at any given time by integrating the payload collection rate (i.e. production rate, described in Section 3.1). Figure 3 demonstrates that $W_{payload}$ provides a good estimate of vertical excavation forces for discrete scraping. For continuous excavation, vertical forces on the cutting surface are small so the vertical forces on the robot overall are again dominated by $W_{payload}$.

Estimating vertical forces on the excavator by $W_{payload}$ simplifies the ensuing analysis by avoiding the need to explicitly model excavation forces. The assumption is valid for the typical classes of planetary excavator presented in this work, and the results are robust to the assumptions made. However, it should be noted that for discrete *scooping* with complex bucket trajectories this simplification should not be assumed to hold without further investigation.

Excavation resistance force is denoted F_{ex} in this work. The non-dimensional ratio of excavation resistance to empty excavator weight is defined as the **excavation resistance coefficient**, and is denoted $\tilde{F} = F_{ex}/W_{robot}$.

2.4. A non-dimensional "lightweight number"

An excavator is operating in the lightweight regime if excavation thrust is less than excavation resistance at any point during digging; in other words, an excavator that produces thrust through traction is in the lightweight regime if at any point

$$P_{20} < F_{ex} \tag{1}$$

Both thrust and resistance depend on the amount of payload collected. The ratio P_{20}/W can be assumed to be constant, as discussed in Section 2.2, and equates to the excavation thrust coefficient, $\tilde{T}$, so

$$P_{20} = \tilde{T}W$$

Note that W is the total weight:

$$W = W_{robot} + W_{payload}$$

Recall that $W_{payload}$ is the weight of currently collected regolith payload, and thus is a quantity that changes (increases) throughout the excavation. Inequality (1) becomes

$$\tilde{T}(W_{robot} + W_{payload}) < F_{ex}$$

Note that F_{ex} is also in general a quantity that changes as a function of the collected regolith payload. Normalizing both sides by the empty robot weight, W_{robot}, leads to

$$\tilde{T}(1 + \tilde{P}) < \tilde{F}$$

Defining a "lightweight number," L, distinguishes the lightweight operating regime when

$$L \equiv \frac{\tilde{T}(1 + \tilde{P})}{\tilde{F}} < 1 \tag{2}$$

The quantitative formulation of the lightweight threshold in $\tilde{F} - \tilde{P}$ space, given above, provides a context for analyzing the suitability of different excavation methodologies for planetary applications. Excavation methods and configurations that are more likely to produce lightweight numbers lower than 1 are in general less suitable than those that are less likely to produce a condition where $L < 1$. The upcoming sections show how L can be characterized for different excavation methodologies, specifically continuous excavation versus discrete scraping, and thus used to make predictions about lightweight operations of robotic excavators that use these methodologies.

3. Characterizing L for continuous and discrete excavation

Characterizing the lightweight number, L, of different excavation methodologies enables examination of their lightweight operation. In particular, characterizing $\tilde{F}$ with respect to changes in payload ratio, that is, $\tilde{F}(\tilde{P})$, enables direct comparison to the lightweight threshold. $\tilde{F}(\tilde{P})$ is a non-dimensional characterization of how excavation resistance forces depend on accumulated/collected regolith payload.

The two classes of excavation methodologies explored in this work are continuous excavation and discrete scraping, as explained in Section 1. Experimental characterizations of $\tilde{F}(\tilde{P})$ for bucket-wheels and flat-plates are described in this section, representing continuous and discrete excavation respectively. The experimental results show why continuous excavation has higher values of L, and thus favorable performance in planetary excavation.

3.1. Experimental setup

Bucket-wheels and flat-plates were constructed of bent 1.6 mm (1/16") thick aluminum plate. Bucket-wheels were made of 12 radially spaced L-shaped blades. Flat-plates were bent to a cutting angle of 10° below the horizontal. Various dimensions of the bucket-wheels and flat-plates are listed in Table 1, with some of the bucket-wheel dimensions explained in Figure 6.

Experiments were conducted in a lunar regolith simulant, GRC-1 (Oravec et al., 2010), contained in a 115 × 68 × 27 cm soil bin. GRC-1 has a bulk density between 1600 kg/m^3 and 1880 kg/m^3, depending on level of compaction. It is a washed industrial silica-sand devoid of agglutinates and of the sub-75 μm basaltic fines that make up 40% of lunar regolith. This enables safe and practical implementation of the large-scale experimentation campaign described in this work, but it also thus does not address the important threats of dust generation and contamination to long-term excavator operation. It also does not directly account for the effect of fines on excavation resistance force. However, as will be pointed out, the results of excavation experiments in GRC-1 are consistent with those conducted by Agui and Wilkinson (2010) in GRC-3, which has a particle size distribution that tracks the full range of lunar regolith particles, including fines (He et al., 2011). The suitability of GRC-1 as a simulant for Martian regolith has not been explicitly characterized.

A consistent soil preparation procedure was followed before each test. First, soil was fully loosened using a gardening spade. The loosened soil surface was made flat by dragging a straight-edged tool across it. To check if the soil surface was horizontal a digital level was placed lengthwise on it. To be accepted as flat horizontal soil the measured angle must be less than 0.5° and the level flush with the soil surface all along its 61 cm length. The soil was modified as necessary until it was sufficiently level, again using the straight-edged tool. The entire surface of the soil was compacted by tamping with a 20 × 20 cm 6.7 kg steel plate with a handle attached to it. Each tamp was achieved by dropping the steel plate from a 5 to 6 cm height; subsequent tamps overlapped each previous one by one-third. The steel plate is one-third of the width of the soil bin so three side-by-side lanes were tamped, for a total width of 60 cm. By executing the compaction procedure along these lanes 10 times it was guaranteed each part of the soil surface would be compacted by at least 10 tamps. This compaction procedure produces reproducible soil strength characteristics in GRC-1, including a penetration soil stiffness of 2.8 ± 0.2 kPa/mm (Diaz Lankenau et al., 2012); equivalent cone index gradients (G) correspond to consistent bulk densities between 1630 kg/m^3 and 1650 kg/m^3 (or 11% to 18% relative density, and thus a very loose to a loose level of compaction; Oravec et al., 2010). Note that subsequent excavator experiments described in Section 4 are conducted in more compacted (medium dense) GRC-1. Bernold (1994) and Goodings and Lin (1995) show that achieving dense (64% to 80% relative density) to very dense (approaching 100% relative density) compaction levels requires substantial and non-linearly increasing effort (in a different, but very relevant, lunar simulant). They also show that the resulting soil strength also increases more than linearly at the high end of the range of compaction levels. Thus it is important to note that the experiments presented in this work do not represent the worst case of excavation resistance forces that may be expected in planetary excavation, especially on the Moon.

To compare bucket-wheels and flat-plates directly, experiments for both excavation methods were run at the same production rate (i.e. payload collection rate). Note that production rate can be estimated by γwdv (where γ is soil density, w is cut width, d is cut depth, and v is advance speed).

Each bucket-wheel and flat-plate test implement was tested at least three times, measuring the force and torque generated during excavation. Test implements were advanced for a distance of 25 cm, using an actuated axis on the soil bin. Before starting each test the test implement was raised above the prepared soil and the load cell was biased. Load cell data collected commenced once test implements were set to the correct depth.

Bucket-wheels advanced perpendicularly to the axis of rotation at a fixed depth and speed through the center of the soil bin. Reaction loads were measured using a 6-DOF force/torque load cell to which the bucket-wheel motor was mounted. The bucket-wheel was then set to rotate and gradually lowered into the soil until the desired depth had been reached. To measure depth a ruler was fixed to the arm holding the bucket-wheel (as seen in Figure 4, left) and vertical distance from the bucket-wheel axle to a 3 mm thick clear plastic plate resting on the soil was measured. Once the bucket-wheel was set to the right depth, the bucket-wheel was set to rotate and advance. As the bucket-wheels did not

Table 1. Scaled dimensions of bucket-wheel (BW) and flat-plate (FP) test implements.

Scale	v	BW D	BW w	BW d	BW e	FP w	FP d	FP l	W_{robot}
1.0	0.07 cm/s	27.6 cm	5.7 cm	2.0 cm	4.6 cm	11.4 cm	1.0 cm	5.8 cm	54 N
1.3	0.10 cm/s	36.8 cm	7.4 cm	2.7 cm	6.2 cm	14.8 cm	1.3 cm	7.5 cm	124 N
1.7	0.13 cm/s	48 cm	9.8 cm	3.5 cm	8.2 cm	19.6 cm	1.7 cm	9.8 cm	283 N
2.3	0.17 cm/s	63 cm	13 cm	4.7 cm	10.9 cm	26 cm	2.3 cm	13.2 cm	645 N

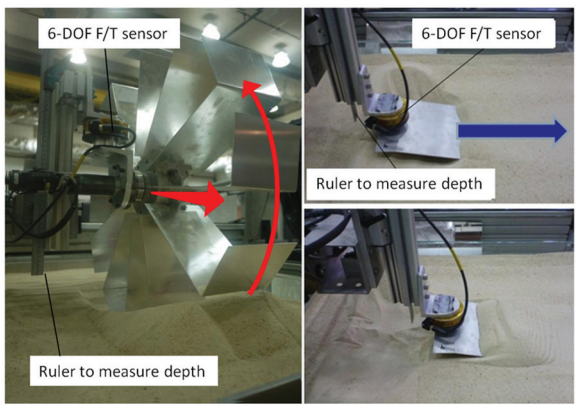

Fig. 4. Experimental setup for measuring excavation resistance forces of bucket-wheels (left) and flat-plates (right) using Force Torque (F/T) sensor.

have side-walls to contain the soil they excavated a shop vacuum cleaner was used to remove soil from each blade just after it fully cleared the soil surface.

Flat-plates were also advanced through the center of the soil bin (Figure 4, right). Reaction loads were again measured using a 6-DOF force/torque load cell to which the angled plate was mounted. The plate was lowered in a space where soil was cleared away to the required depth. Depth was measured similarly to the procedure for bucket-wheels. Once the flat-plate was set to the right depth, it was advanced through the soil.

The lack of side-walls has a negligible effect on excavation resistance for the flat-plates in these experiments, but does affect the measured resistance for bucket-wheels. Osman explains that a cutting blade wider than six times the cutting depth can be considered as infinitely wide (Osman, 1964). For the flat-plate experiments, the blade width is more than 11 times the depth, as shown in Table 1. For the bucket-wheel experiments, though, width is less than three times the depth which can introduce some error. In this case, as soil may be pushed to the sides, one expects measured excavation resistance to be lower than if the sides were closed; on the other hand, soil that has fallen out the sides as each bucket exits the terrain increases resistance for the subsequent bucket. The analysis in the next subsection will demonstrate that the key results hold even if bucket-wheel forces were underestimated by a full order of magnitude, which is extremely implausible. Experiments with closed-side excavator buckets in the subsequent section also further verify the result.

3.2. Experimental results and their relation to L

Experimental results imply higher values of L for continuous excavation than for discrete scraper excavation.

Typical results of experiments comparing bucket-wheel and flat-plate excavation forces are shown in Figure 5. Raw force/torque data, collected as described in the previous subsection, provides horizontal excavation resistance force (i.e. F_{ex}) versus time. However, because the bucket-wheel and flat-plate operate at the same production rate, this can be easily converted to F_{ex} versus payload accumulated/collected (by multiplying the x-axis scale by the production rate). Furthermore, continuous and discrete excavators of equal weight are compared, for the sake of a fair comparison. This allows normalization of the axes, for bucket-wheels as well as for flat-plates, by a single robot weight W_{robot}, to arrive at the non-dimensional axes $\tilde{F}$ and $\tilde{P}$ shown in the figure.

The particular value of W_{robot} for any set of experiments was estimated by comparison to excavator prototypes and existing extraterrestrial rovers. Specifically, it is estimated that an excavator with an 80 cm diameter bucket-wheel would require a robot with an empty mass of at least 150 kg. This can be compared to the 312 kg Scarab rover equipped with an 80 cm diameter bucket-wheel (see Figure 9) or the 200 kg Polaris excavator presented in Section 5; it is assumed that such terrestrial prototypes could be further optimized for mass. Another comparison is to the the NASA Mars Exploration Rovers, which have interior chassis widths of 1 m and a mass of 180 kg. Upcoming analysis in this section will show why this estimate of robot mass is intentionally conservatively low. The value of W_{robot} for each set of experiments was scaled from this example according to the method described in Section 3.3.

One observation that can be drawn from these experimental results in Figure 5 is that $\tilde{F}(\tilde{P})$ for both continuous excavation and discrete scraping can be represented by a linear approximation. Straight-line approximations are included in the figure for illustration. Agui and Wilkinson (2010) also found that, equivalently, $F_{ex}(x)$ can be approximated by straight lines for discrete scraping.

This linear approximation can be denoted by

$$\tilde{F} = \tilde{F}_0 + \tilde{F}'\tilde{P}$$

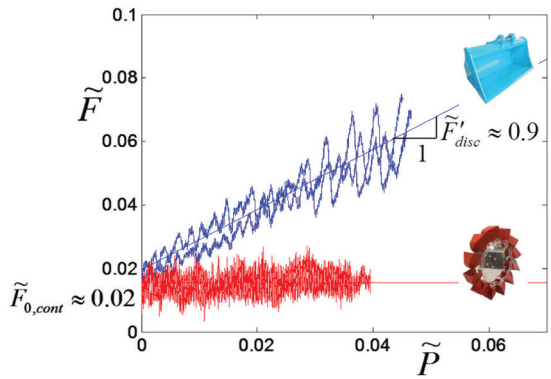

Fig. 5. Excavation resistance coefficient, $\tilde{F}$, vs payload ratio accumulated, $\tilde{P}$, for continuous excavation (red) and discrete scraping (blue) at equal production rates. Raw data and linear regression are shown. $\tilde{F}' \approx 0.9$ for discrete scraping, which is substantially higher than both $\tilde{F}_{0,disc}$ and $\tilde{F}_{0,cont}$; as a result, continuous excavators achieve higher lightweight numbers, L, and are thus more suitable for lightweight operation.

with $\tilde{F}'$ being shorthand for $d\tilde{F}/d\tilde{P}$, the slope of the line. Factoring out $\tilde{F}_0$ and substituting into (2) leads to

$$L = \frac{\tilde{T}(1+\tilde{P})}{\tilde{F}_0(1+\frac{\tilde{F}'}{\tilde{F}_0}\tilde{P})} \tag{3}$$

Another observation is that $\tilde{F}'$ is zero for continuous excavation and positive for discrete scraping. This increasing discrete excavation resistance force can be attributed to the ever-growing accumulation of soil at the cutting edge as a scraping bucket is filled. This soil at the cutting edge resists entry of further soil into the bucket. For continuous excavators, on the other hand, excavation resistance force is bounded. When each individual bucket has accumulated an appreciable amount of soil it clears the ground and the next, soil-free, bucket has already started cutting.

The upshot is that accumulation of payload has opposite effects on L for continuous and discrete excavators. The most challenging operating point for continuous excavators is at the start of a cut, after which digging gets easier. For discrete excavators on the other hand, digging becomes more difficult as cutting proceeds.

For continuous excavation, $\tilde{F}$ is bounded so a linear approximation yields $\tilde{F}'_{cont} = 0$. Substituting into (3) gives

$$L_{cont} = \frac{\tilde{T}(1+\tilde{P})}{\tilde{F}_{0,cont}}$$

In this case, adding payload (i.e. increasing $\tilde{P}$) increases L. Additional payload pushes continuous excavators away from the lightweight regime:

$$\min(L_{cont}) = \frac{\tilde{T}}{\tilde{F}_{0,cont}} \tag{4}$$

For discrete scraping, L decreases with increasing $\tilde{P}$ as long as $\tilde{F}'/\tilde{F}_0 > 1$. In reality, $\tilde{F}'_{disc}$ is greater than $\tilde{F}_{0,disc}$ by between one and two orders of magnitude (see Figure 5; similar results can be derived from the data in Agui and Wilkinson, 2010). This means that additional payload pushes excavators towards the lightweight regime. The theoretical lower bound (as $\tilde{P} \to \infty$) for discrete L approaches

$$\min(L_{disc}) = \frac{\tilde{T}}{\tilde{F}'_{disc}}$$

Even with an assumption limiting payload ratio to $\tilde{P} = 0.5$ (which is more realistic, as discussed in Section 2.1),

$$\min(L_{disc}) < \frac{3\tilde{T}}{\tilde{F}'_{disc}} \tag{5}$$

Finally, one other observation that can be drawn from Figure 5 is that $\tilde{F}'_{disc}$ is orders of magnitude greater than $\tilde{F}_{0,cont}$, at equivalent production rates. Meanwhile, comparing equation (4) to equation (5), a factor of just 3 ensures a higher minimum lightweight number, $min(L)$, for the continuous configuration (for the practical case of $\tilde{P} = 0.5$). This higher margin on L means continuous excavators can operate more safely with less risk of crossing into the lightweight regime than discrete scraper excavators.

The result that $\tilde{F}'_{disc}$ is substantially greater than $\tilde{F}_{0,cont}$ is not entirely independent of W_{robot}. $\tilde{F}_0$ is dependent on W_{robot} while $\tilde{F}'$ is not (it cancels out of the ratio $\tilde{F}'/\tilde{P}$). This is why a conservatively low estimate was made for W_{robot}; any larger value only increases the factor by which $\tilde{F}'_{disc}$ is greater than $\tilde{F}_{0,cont}$. In fact, due to the large magnitude of the difference, even an order of magnitude *lower* W_{robot} (which is totally unrealistic) would not change the result observed here. The generality of these results is supported by additional experiments in subsequent sections, and will be discussed explicitly in the conclusion.

3.3. *Testing bucket-wheels and flat-plates at various scales*

To study whether the advantages of continuous excavation over discrete scraping are sensitive to the scale of excavator, four sizes each of bucket-wheels and flat-plates were tested. All aspects of the test implements with a length dimension were scaled proportionally. For bucket-wheels this includes wheel diameter D (A), blade width w (B), blade tangent extension e (C), cut depth, and advance speed. Angular speed, number of blades, and blade thickness were kept constant. A, B, and C are shown in Figure 6. For flat-plate tests, scaled parameters include plate width, plate length l, and advance speed (blade thickness again kept constant). Dimensions for bucket-wheels and flat-plates are summarized in Table 1.

Again note that production rate, which can be estimated by γwdv (where γ is soil density, w is cut width, d is cut depth, and v is advance speed), is consistent between bucket-wheels and flat-plates of the same scale. In fact,

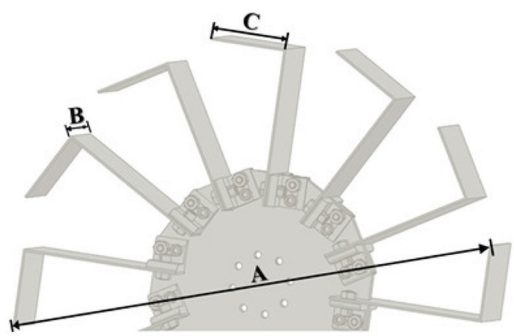

Fig. 6. Bucket-wheel dimensions: wheel diameter D (A), blade width w (B), and blade tangent extension e (C).

the scaling ensures that production ratio (production rate normalized by a representative excavator mass) is constant among all tests.

The scaling of advance speed and bucket-wheel diameter, with angular rate kept constant, results in a tangential bucket-wheel cutting speed five times the advance speed during each test. Cutting speeds were low enough that no dynamic effects needed to be considered. These values never exceeded 1 cm/s (measured as the sum of advance speed plus tangential speed).

Bucket-wheel and flat-plate excavation resistance was measured at four separate scales, with these scales defined in Table 1. Each bucket-wheel and flat-plate was tested at least three times, measuring the force and torque generated during excavation as described in Section 3.1. To non-dimensionalize, as described in Section 3.2, test data was normalized by a W_{robot} for each scale, as listed in Table 1. The values of W_{robot} are based on scaling from a 150 kg excavator with an 80 cm bucket-wheel, as described in the previous section; note that the weights are scaled according to the cube of length dimensions, because robot mass scales with volume.

Figure 7 shows experimental results from bucket-wheel and flat-plate tests at each of the four scales. Raw data from multiple runs at the same scale are overlaid with a linear regression of combined test data. The data shows that at all scales bucket-wheel excavation resistance is bounded, while it rises approximately linearly with accumulating payload for flat-plate excavation. Variability around this linear rise also tends to increase as a flat-plate cut proceeds. Crucially, to add support to the conclusions drawn in the previous section, $\tilde{F}'_{disc}$ is greater than $\tilde{F}_{0,cont}$ by a factor of at least 40 to 60 regardless of scale (with no notable trend in this factor with increasing scale). Thus, spanning various scales of excavation robots, continuous excavation is more suitable for planetary operations than discrete scraping. This is summarized in the archetypal diagram in Figure 8.

4. Gravity-offloaded excavator experiments

The benchtop experiments described in the previous section suggest that continuous excavators should perform better in reduced-gravity conditions than discrete scrapers. This section describes experiments, utilizing a full robotic system (i.e. combining both mobility and excavation subsystems), that are consistent with this prediction. The experiments themselves were recently reported by the authors (Skonieczny et al., 2015). Section 4.1 summarizes the setup of these experiments. The major extension of this section is Section 4.2, which situates and discusses the experiments within the broader context of the theoretical framework that has now been developed within this work. Section 4.3 also extends beyond the prior work (Skonieczny et al., 2015) with a substantially more thorough treatment of the results.

The experiments described here contribute a first result of studying excavators digging while subject to gravity offload. Five-sixths of the robot's weight is "offloaded," in this case borne by vertical cable tension, and the remaining one-sixth of the robot's weight loads the soil via its wheels. This mimics the robot's weight on its wheels on the Moon. There are two methods for "offloading" a robot's weight. One method is to make a full geometric scale robot that only has the proportion of mass corresponding to the gravity at its target destination. An example is NASA JPL's "SCARECROW" rover (White et al., 2008), whose 3/8 mass loads the wheels equivalently to the full mass Curiosity rover on Mars. For lunar rovers, however, it would be very difficult to produce a full-scale rover with only one-sixth mass, even if only suspension and mobility subsystems were included. The other offloading method is to apply an upward force to the robot to offset part of its weight; this is the method used in this research.

In actual reduced gravity, the regolith is also subject to reduced self-weight which affects its interactions with the wheels and cutting tools. Short of testing on the surfaces of the Moon and Mars directly, reduced-gravity flights (Boles et al., 1997; Kobayashi et al., 2010) offer the testing environment that is most representative of the reduced gravity in planetary operations. In these tests, the regolith and robot are both subject to reduced g. Future research on lightweight excavation would benefit from testing in reduced-gravity flights. Opportunities for such tests are infrequent, though, and their scale (both spatially and temporally) is severely constrained by the logistics of the flights.

Gravity offload cannot capture the effects of reduced gravity on regolith. Reasoning presented below shows that gravity offload nonetheless provides a more balanced indication of performance in reduced g than simply testing a full mass excavator in Earth gravity. Testing only in Earth gravity with a full mass excavator over-predicts performance relative to reduced gravity, introducing the risk of unforeseen performance losses or failures during planetary operations.

Using the terminology introduced in Sections 2.2 and 2.3, operation in reduced gravity reduces the excavation thrust coefficient, $\tilde{T}$, while also typically increasing the

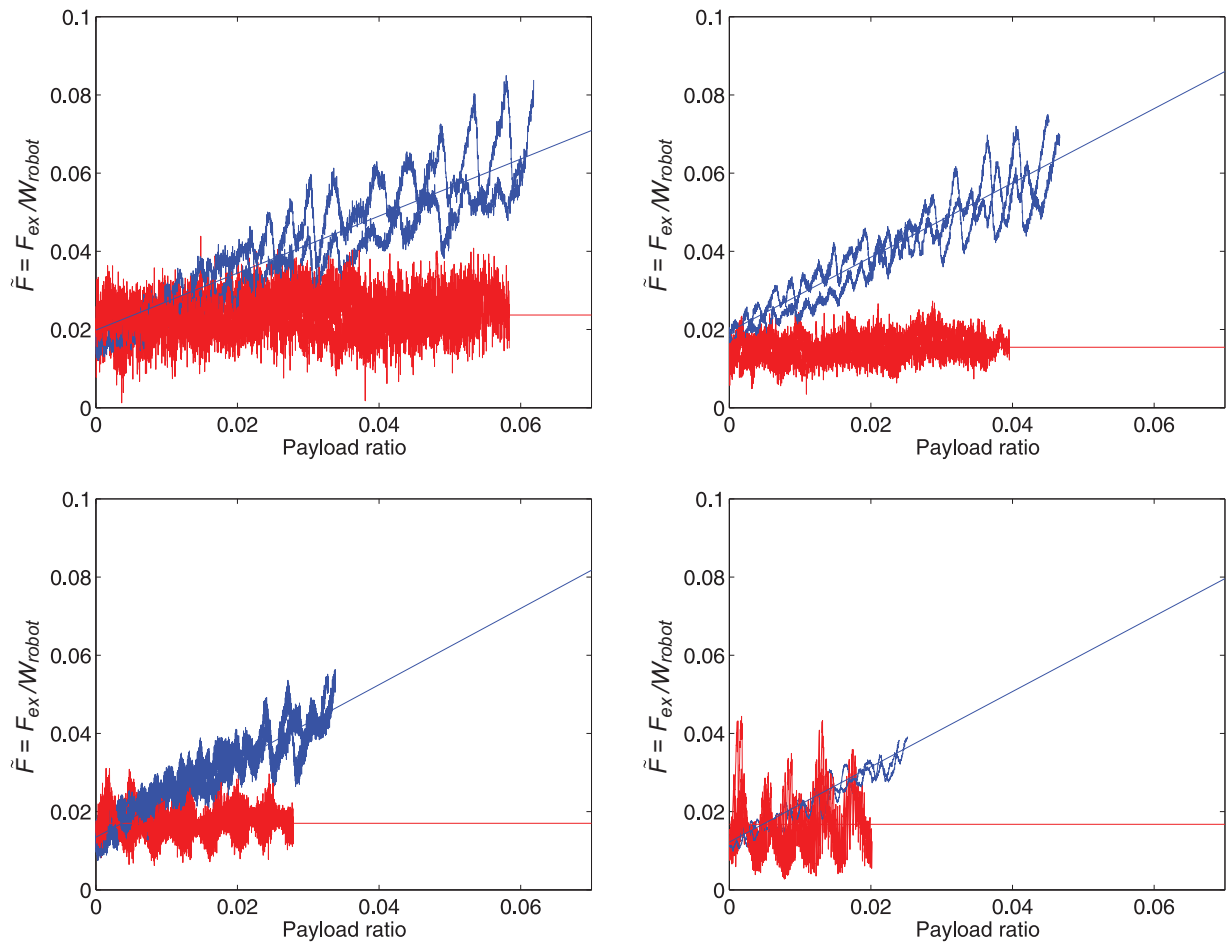

Fig. 7. Discrete flat-plate excavation resistance (blue) increases without bound, while continuous bucket-wheel excavation resistance (red) remains low and bounded. Plots show excavation resistance coefficient, $\tilde{F}$, vs payload ratio accumulated, $\tilde{P}$, in cohesionless soil at four different scales: 1.0 (top left), 1.3 (top right), 1.7 (bottom left), and 2.3 (bottom right). Raw data and linear regression are shown.

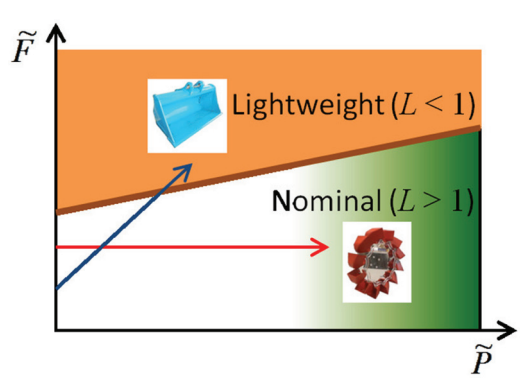

Fig. 8. At equivalent production, continuous excavators are less likely to cross the lightweight threshold. This is equivalent to continuous excavators having higher values of L. [now referenced at end of Section 3.3]

excavation resistance coefficient, $\tilde{F}$. This translates into reduced relative mobility capabilities combined with relatively more challenging digging tasks when compared to operation on Earth. Testing in Earth gravity is equivalent to assuming these coefficients will instead stay constant, thus over-predicting performance on both accounts. Gravity offload testing, on the other hand, is equivalent to again assuming that $\tilde{T}$ stays constant (again over-predicting mobility performance) but that $\tilde{F}$ increases according to the worst-case scenario (thus under-predicting the actual excavation performance in reduced g).

As explained in Section 2.2, $\tilde{T}$ stays constant if the vertical load (represented by the weight, W) on the wheels changes but geometric scale and gravity are kept constant. This is the case with gravity offload. This is because both thrust, T, and resistance, R, are reduced under this type of lowered W. Thrust is reduced because of reduced traction from frictional shearing, which is proportional to loading. Resistance is reduced because lower load in the same gravity causes less sinkage into regolith, and sinkage contributes prominently to rolling resistance (Bekker, 1969). If R is assumed proportional here to W to a first-order approximation, and we recall that drawbar pull is net traction (i.e. $T - R$), we can readily see why $\tilde{T}$ would remain approximately constant in this situation:

$$\tilde{T} = \frac{P_{20}}{W} = \frac{T - R}{W}$$

On the other hand, if W changes *due to a corresponding reduction in gravity*, then $\tilde{T}$ does not stay constant but rather *reduces*. Kobayashi's reduced-gravity parabolic flights showed that wheel sinkage is *not* reduced driving in lower gravity (Kobayashi et al., 2010), meaning R is not reduced; frictional loading (and thus T), though, is again reduced in this form of W reduction. This imbalance leads to the reduction of $\tilde{T}$ in this situation.

To see that reduced gravity not only reduces excavation thrust coefficient, $\tilde{T}$, but also increases excavation resistance coefficient, $\tilde{F}$, we re-examine another set of experimental results, this time produced by Boles et al. Their work compared excavation resistance forces measured in Earth gravity to ones measured during reduced g flights, in otherwise identical experiments. The results were that the ratio of excavation resistance in 1/6 g to excavation resistance in 1 g, $F_{ex,1/6g}/F_{ex,1g}$, could fall anywhere between one-sixth (i.e. reduction in resistance proportional to reduction in gravity) and one (i.e. no reduction in resistance at all despite reduction in gravity) and was typically one-third in their particular experiments (Boles et al., 1997). Zacny et al. (2010) analytically showed that the scaling of excavation forces in reduced gravity depends on the cohesiveness of regolith and the scale of the robot, with a small-scale robot in highly cohesive regolith resulting in the least relative reduction of resistance. This particular disadvantageous case is a likely one for lunar ISRU. Excavation resistance not scaling down at all with reduction in gravity corresponds to the case of $F_{ex,1/6g}/F_{ex,1g} = 1$ described by Boles et al, and would lead to a six-fold increase in $\tilde{F}$ (recall its definition: F_{ex}/W) in 1/6 g.

Gravity offload reduces robot weight but not regolith weight, and this has no effect on longitudinal soil–tool forces. F_{ex} thus remains constant regardless of whether the robot is offloaded or not, making gravity offload directly representative to the worst case $F_{ex,1/6g}/F_{ex,1g} = 1$ described above. Neither lunar nor Martian regolith is in fact purely cohesive, so this type of worst-case experiment does overestimate how poorly excavation resistance forces would be expected to scale on these planetary surfaces.

In summary, excavating with gravity offload underestimates detrimental effects of gravity on traction, but overestimates detrimental effects on excavation resistance. Although not ideal, this is more balanced than excavating in Earth gravity which underestimates detrimental effects on both traction *and* resistance. One of the contributions of these first-of-their-kind gravity-offloaded excavation experiments, as described by Skonieczny et al. (2015) and here, is thus a more realistic and conservative characterization of expected planetary excavator performance.

4.1. Experimental setup

To explore the differences in lightweight thresholds for continuous and discrete excavation, gravity-offloaded excavation experiments were conducted at NASA Glenn Research Center's (GRC) Simulated Lunar Operations (SLOPE) lab. A large soil bin with GRC-1 (Oravec et al., 2010) lunar simulant was used (GRC-1 is described in Section 3.1). Carnegie Mellon University's Scarab robot (for a detailed description, see Bartlett et al., 2008; Wettergreen et al., 2010) was used with either continuous bucket-wheel or discrete bucket excavation tools mounted to its structural chassis.

This research program developed a gravity offload apparatus for the SLOPE lab, as shown in Figure 9. Experimental setup parameters are summarized in Table 2. Additional setup and soil preparation details can be found in Skonieczny et al. (2015).

4.2. Predicted lightweight numbers

Estimates of lightweight numbers, L, for the various test conditions predict that Scarab will not enter the lightweight regime for either mode of excavation at 1 g, but that it will cross the lightweight threshold when performing discrete scraping under gravity offload.

A maximum payload ratio of $\tilde{P} = 0.5$ is assumed for the excavation configurations of interest. In this case, equation (5) is applicable to discrete scraping excavation, and equation (4) applies to continuous excavation (independent of the choice of $\tilde{P}$). Excavation thrust, $\tilde{T}$, is the same for both modes of excavation, as the excavator uses the same "spring tires" for all the experiments. For these tires, $\tilde{T} \approx 0.25$, as seen in Figure 2. The value of $\tilde{F}'_{disc}$, between 0.2 and 0.3, is estimated from experiments with a similar discrete bucket (Agui and Wilkinson, 2010). Force measurements from preliminary tests with the bucket-wheel suggest a $\tilde{F}_{0,cont}$ between 0.002 and 0.004. These values lead to estimates of L between 60 and 130 for continuous excavation, and between 2 and 4 for discrete scraping, in 1 g.

Gravity offload reduces drawbar pull proportionally to the reduction in weight, without changing excavation resistance. This effectively reduces L in step with the weight reduction, leading to L-values of between 10 and 20 for continuous excavation, and between 0.3 and 0.7 for discrete scraping when operating with the excavator weight offloaded to one-sixth. An L below 1 for the discrete scraping case represents crossing the lightweight threshold and thus an inability to safely collect a payload ratio of 0.5. Table 3 summarizes these predictions.

4.3. Experimental results

Experiments show that in 1 g neither continuous nor discrete excavation crosses the lightweight threshold. On the other hand, in gravity-offloaded 1/6 g, continuous excavation is again productive but discrete scraping crosses the lightweight threshold and collects too little regolith payload to be productive.

Rover speed was measured for each test condition, including baseline tests of driving without digging. Speed

Table 2. Summary of experimental parameters for gravity-offloaded continuous and discrete excavation.

Parameter	Continuous excavation	Discrete scraping
Regolith simulant Bulk density Relative density (Oravec et al., 2010)	GRC-1 1700 kg/m^3 to 1740 kg/m^3 36% to 50% (medium dense)	GRC-1 1700 kg/m^3 to 1740 kg/m^3 36% to 50% (medium dense)
Nominal production rate	0.5 kg/s	0.5 kg/s
Tool geometry	Bucket-wheel diameter: 80 cm Cutting depth: 5 cm Number of buckets: 12 Bucket width: 15 cm	Bucket width: 66 cm Cutting depth: 2 cm Cutting angle: 15° below horizontal
Trajectory Commanded robot speed Speed control for offload assembly Robot position tracking	Straight line (directly below hoist rail) 2.7 cm/s Open-loop Laser total station (1 Hz)	Straight line (directly below hoist rail) 2.7 cm/s Open-loop (1 g) Manual (offloaded to $1/6$ g) Laser total station (1 Hz)
Scarab mass Weight distribution	312 kg 54% on rear wheels	312 kg 54% on rear wheels
Instrumentation	Two-axis inclinometer (cable angle) Single-axis load cell (cable tension) Six-axis force/torque (excavation forces)	Two-axis inclinometer (cable angle) Single-angle load cell (cable tension)

Fig. 9. Gravity offload testing with bucket-wheel (left) and front-loader bucket (right) on the Scarab robot. A cable pulls up on the robot, tensioned by weights acting through a 2:1 lever arm. The weights and lever assembly hang from a hoist that is pulled along a passive rail by a separate winch-driven cable (Skonieczny et al., 2015).

was calculated based on analyzed total station data. Each test condition was run three or four times. Figure 10 shows rover speed results for excavating in 1 g and offloaded $1/6$ g. Forward progress is maintained in all conditions except discrete scraping with gravity offload. Means and standard deviations of the speed data are given in Table 4. Slightly slower mean speeds are observed in 1 g, because the compliant "spring tires" are compressed to a smaller wheel radius under the greater weight. Excavation and gravity offload both introduce a small amount of additional variability in speed compared to driving without digging in 1 g.

During actual operations, one would expect the discrete scraper bucket to be actuated to curl out of the regolith prior to the vehicle stalling. Although this control maneuver could improve operational safety, it does not change the fact that the discrete scraper falls well short of achieving the high payload ratio required for productivity.

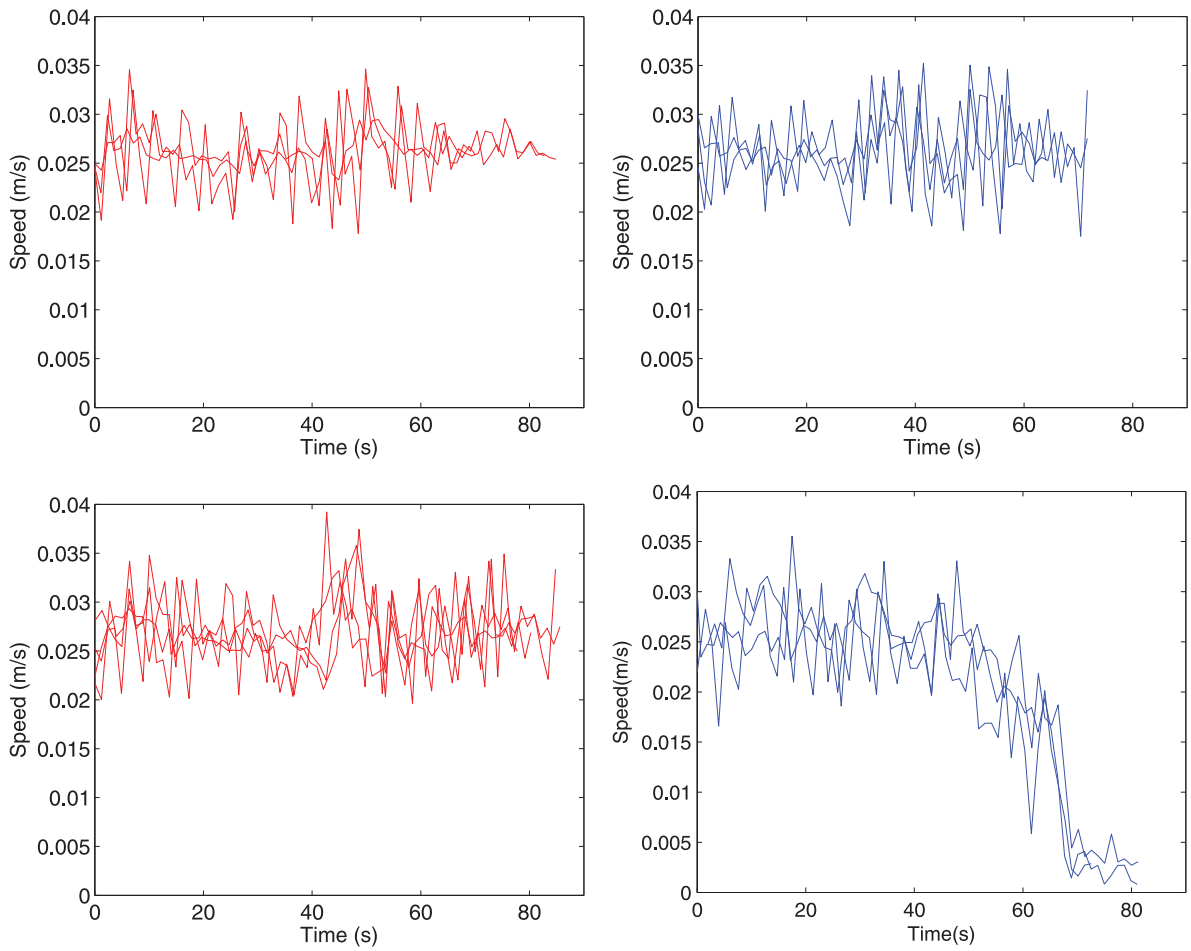

Fig. 10. Excavator forward driving speed during continuous excavation in 1 g (top left), discrete scraping in 1 g (top right), continuous excavation in gravity-offloaded $1/6$ g (bottom left), and discrete scraping in gravity-offloaded $1/6$ g (bottom right; time axes aligned at stall point). The excavator maintains constant forward progress in all cases except discrete scraping with gravity offload (Skonieczny et al., 2015).

Table 3. Predicted values for L for the relevant experimental conditions. The excavator is predicted to cross the lightweight threshold ($L < 1$) when performing discrete scraping under gravity offload.

	1 g	Offloaded to $1/6$ g
Continuous excavation	60–130	10–20
Discrete scraping	2–4	0.3–0.7

Figure 11 shows the regolith collected in the discrete bucket after 1 g and offloaded $1/6$ g experiments. Discrete scraping in gravity-offloaded $1/6$ g collected only 15 to 20 kg before vehicle stall. In contrast, approximately 45 kg was collected in 1 g experiments; this same mass was also collected in all the 1 g and offloaded $1/6$ g continuous excavation experiments. It is crucial to note that the 15 to 20 kg of payload collected by discrete scraping in gravity-offloaded $1/6$ g, a final payload ratio less than 0.07, severely limits the potential productivity of such a system.

Table 4. Discrete scraping offloaded to $1/6$ g is the only test condition that does not maintain constant steady state (S/S) velocity throughout. Note that σ_v represents the mean of the three tests' σ values, not the σ of the three tests' mean velocities (which showed negligible variation between tests of any single set) (Skonieczny et al., 2015).

Excavation type	"Gravity"	Average v	σ_v
Driving only	1 g	2.6 cm/s	0.2 cm/s
Continuous	1 g	2.6 cm/s	0.3 cm/s
Discrete	1 g	2.6 cm/s	0.4 cm/s
Driving only	$1/6$ g	2.7 cm/s	0.3 cm/s
Continuous	$1/6$ g	2.7 cm/s	0.3 cm/s
Discrete	$1/6$ g	no S/S	n/a

It is important to consider any potential effects the gravity offload apparatus may have on $\tilde{T}$ or $\tilde{F}$ during excavation testing, either by inadequately maintaining consistent W or by inducing longitudinal (and/or lateral) forces. Figure 12

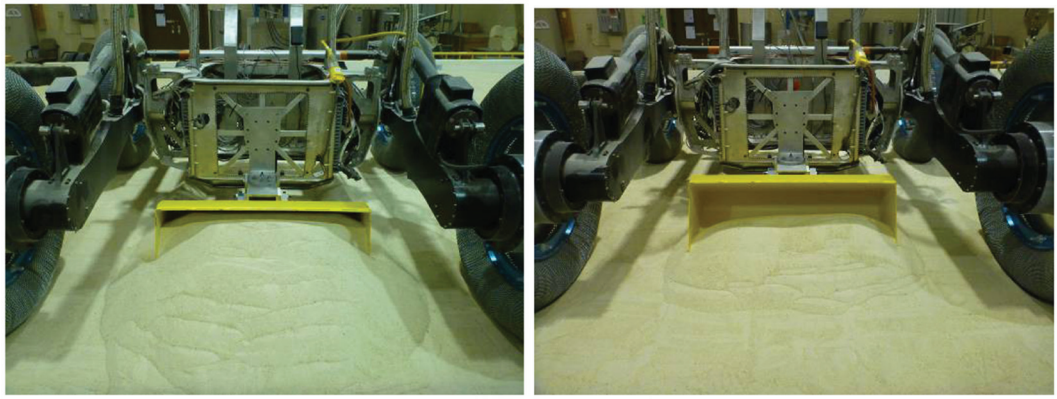

Fig. 11. Discrete bucket at end of excavation tests without gravity offload (left; full with approximately 45 kg of GRC-1) and with gravity offload (right; only 15 to 20 kg of GRC-1 collected).

shows longitudinal cable angle and cable tension for a continuous excavation test. Cable angle was unbiased about the vertical, with a mean absolute value of just 0.1°. Transient motions of the cable did not exceed 0.8° from the vertical for more than a fraction of a second; with a cable tension of 2600 N, this corresponds to brief transients of 35 N, or 7% of the offloaded excavator weight. Cable tension varies by just ±1% which, amplified by the offloading ratio, corresponds to a 5% variation in the offloaded excavator weight. Figure 13 shows the longitudinal cable angle and cable tension for a discrete excavation test; variability in angle and tension were again small. These results demonstrate the quality of gravity offload achieved in this experimentation campaign.

A transverse bucket-wheel also induces lateral excavation forces that must be accounted for. These forces are transferred to the wheels, leading either to lateral reaction forces that balance them or to vehicle slew and/or yaw. Lateral forces can be reacted by friction between the wheels and soil. For wheels with lateral gripping features that capture soil particles, the soil strength beneath the wheel can also react lateral forces. Finally, wheels experiencing sinkage can react against the soil with their sidewalls. If the lateral forces are high enough to induce significant lateral soil shearing, the vehicle will depart from its nominal forward trajectory.

During gravity-offloaded bucket-wheel excavation tests, excavation resistance forces were low enough that mobility was unaffected. Force data was collected using a six-axis force/torque sensor mounted between the bucket-wheel actuator and the robot chassis. Lateral and longitudinal forces acting on the bucket-wheel were on the order of 10 to 20 N and 5 to 10 N, respectively (as shown in Figure 14). Lateral forces of this magnitude are less than 5% of the vehicle's offloaded weight, and did not cause discernible slew or yaw in the robot's trajectory.

The gravity-offloaded excavation experiments described here directly demonstrate the advantages of continuous excavation over discrete scraping in lightweight conditions.

Table 5. Summary of engineering specifications of the Polaris excavator robot.

Mass	200 kg
Nominal payload capacity	80 kg
Payload ratio	40%
Top driving speed	40 cm/s
Dump height	50 cm

5. Development of a planetary bucket-wheel excavator prototype

With some of the key advantages of continuous excavation now established, this section describes a continuous planetary excavator prototype that addresses some practical implementation considerations for planetary environments. The Polaris excavator, shown in Figure 15, is intended for productive ISRU. Key specifications are listed in Table 5. Note that prior research by the authors has already shown that high payload ratio and driving speed govern productivity (Skonieczny et al., 2014). The additional practical consideration featured here is novel direct transfer from bucket-wheel to collection bin.

Transferring regolith from bucket-wheel to collection bin presented a serious challenge for past planetary bucket-wheel excavator prototypes, and as a result bucket-ladders have gained favor (Johnson and Van Susante, 2006). These bucket-ladder configurations use chains to move buckets along easily shapeable paths, making transfer to a collection bin easy. During the NASA Regolith Excavation Challenge and subsequent Lunabotics mining competitions (which require digging lunar regolith simulant for 30 minutes, and highly prized productivity), the winners all employed bucket-ladders driven by exposed chains (Mueller and Van Susante, 2011). However, bucket-ladder chains are exposed directly to the soil surface and could degrade very quickly in harsh lunar regolith and vacuum. The abrasiveness of lunar regolith rapidly degrades exposed sliding contacts or flexible materials (Stubbs et al., 2005;

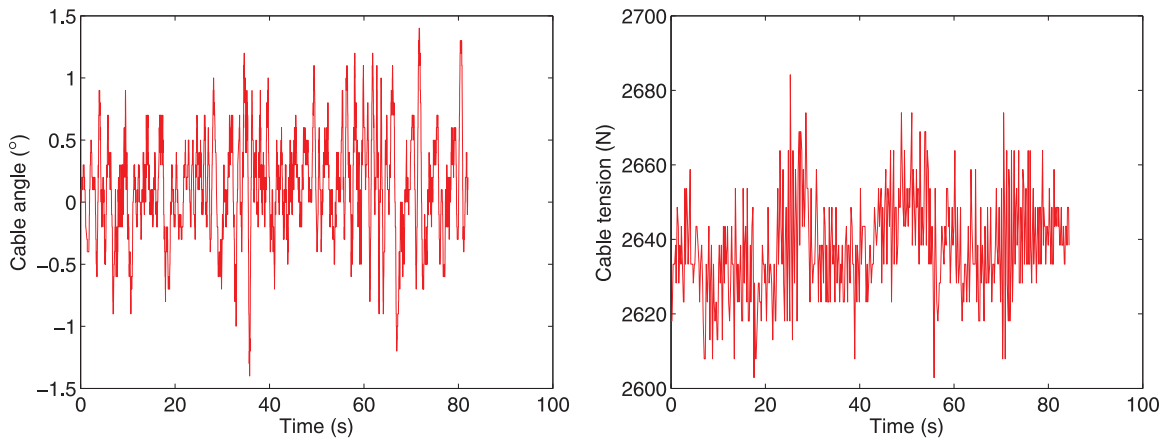

Fig. 12. Longitudinal angle (left) and tension (right) of the gravity offloading cable during a continuous excavation experiment, showing minimal variation.

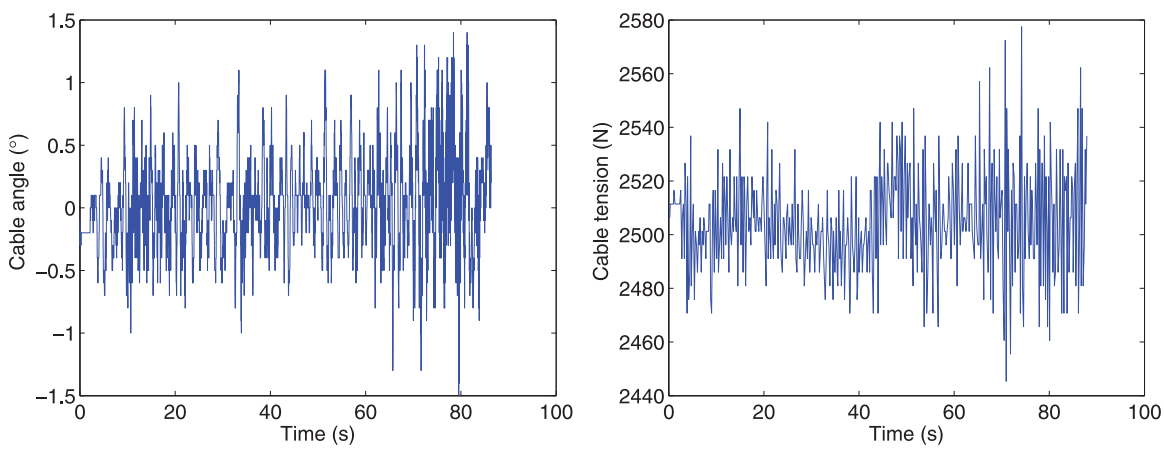

Fig. 13. Longitudinal angle (left) and tension (right) of the gravity offloading cable during a discrete excavation experiment, showing minimal variation (Skonieczny et al., 2015).

Harrison et al., 2008). Exposed bucket-ladder chains may not be relevant to operation in lunar conditions based on these concerns.

Dust poses a risk to excavation equipment (Slane, 1994), and must be considered both in terms of the amount of dust generated and the ability to seal components from contamination. The multiple cutting and transfer points of both bucket-ladders and bucket-wheels risk increasing dust generation during excavation and material transfer relative to discrete excavators, which is an important consideration to trade off against the advantages of continuous excavation described in this work. All excavators (continuous or discrete) generate hazardous dust when dumping their regolith payloads, though, so dust generation considerations are not enough to dismiss continuous excavators altogether. Within the class of continuous excavators, bucket-ladders pose greater contamination risk than prudently configured bucket-wheels.

The novel excavator configuration presented here has a bucket-wheel, mounted centrally and transverse to the driving direction, that directly transfers regolith into a collection bin. Regolith is carried to the top of the wheel by each individual bucket, and then drops down through the inner openings of the buckets into the collection bin as shown in Figure 16. The bucket-wheel is a single moving part, with no need for chains or conveyors, reducing complexity and risk of contamination from regolith and dust. It should be noted, however, that the excavator testing described here was completed in GRC-1 so dust generation and contamination tolerance could not be directly characterized. This configuration offers a solution to the transfer problem for bucket-wheels identified in past literature by eliminating the need for additional transfer mechanisms entirely.

As already shown in Figure 14, transverse bucket-wheel orientation does not induce lateral excavation forces large enough to slew or yaw the robot while five-sixths gravity-offloaded. Preliminary experiments show spikes in both lateral and longitudinal excavation force when encountering rocks; such force spikes, depending on their frequency and severity, could impact the productivity of either continuous or discrete excavation. Polaris' bucket-wheel features a design that reduces opportunities for rocks to get stuck in

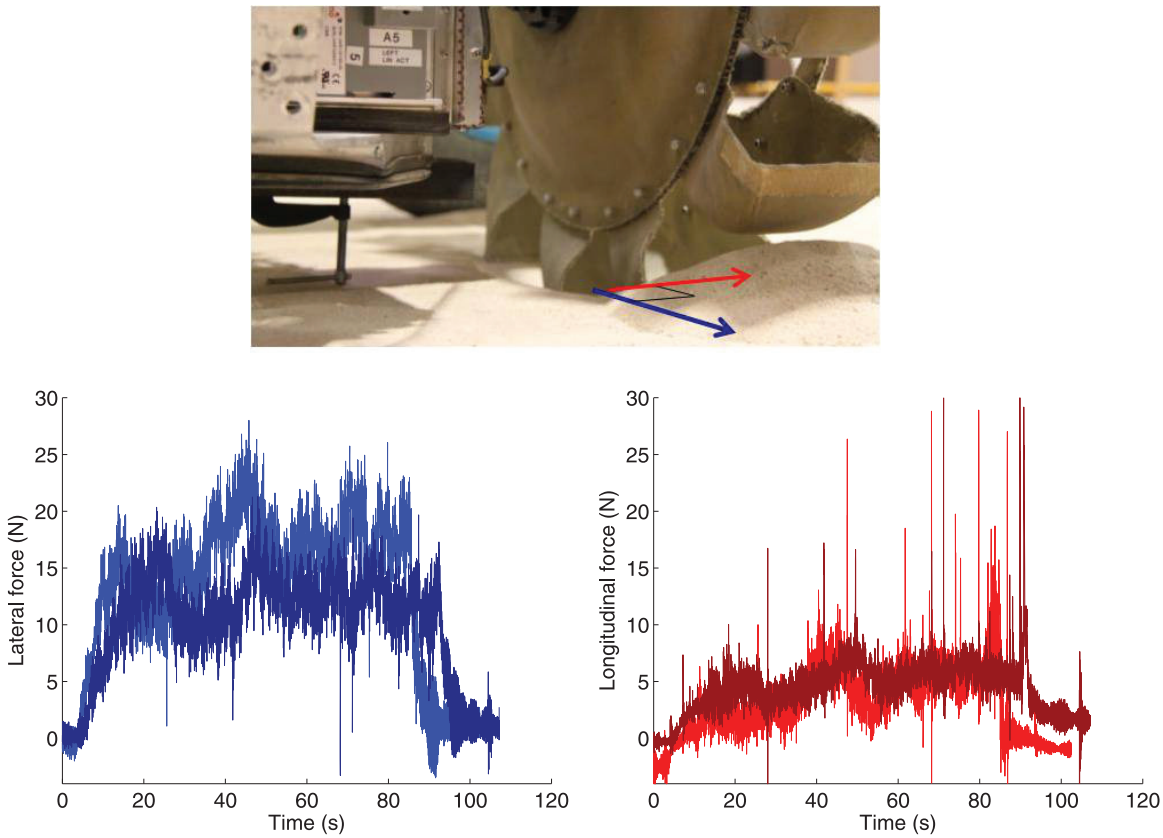

Fig. 14. Lateral (lower left) and longitudinal (lower right) excavation resistance forces are low enough that excavator mobility is unaffected, even with gravity offload. Lateral forces, due to the transverse bucket-wheel orientation, do not cause the excavator to slew or yaw. Lateral (blue) and longitudinal (red) directions relative to the bucket-wheel are indicated in the upper image.

Fig. 15. Polaris excavator featuring continuous bucket-wheel excavation and high payload ratio.

the buckets: the gaps between the bucket openings and the structure of the buckets ahead are smaller than the smallest gap within the bucket itself. In preliminary testing, small rocks passed through the bucket-wheel into the collection bin while larger ones were pushed aside. Further study of excavation in the presence of rocks is suggested as a future direction of work.

Polaris has demonstrated mining productivity by excavating and delivering 1040 kg in 58 minutes, with an average round-trip distance of approximately 14 m. This was demonstrated in GRC-1 simulant at NASA GRC's SLOPE lab. Approximately one-third of the operation time was spent digging in the simulant, and 17 dig-dump task cycles were executed. The robot was teleoperated. Full-scale excavation tasks such as this one, but conducted with gravity offload, are suggested for future work.

6. Conclusions and future work

This research shows that continuous excavation is more suitable for planetary robotic ISRU operations than discrete scraping. This conclusion is principally drawn from an analytical framework introduced here, defining a lightweight number, L; L greater than 1 indicates nominal operations while L less than 1 indicates a regime where excavation resistance forces overwhelm traction forces and wheel slip exceeds 20%. Continuous excavators are shown to typically have significantly higher values of L than discrete scrapers. The primary conclusion is also supported by experiments that directly compare continuous and discrete excavation in lightweight conditions; when offloaded by five-sixths weight (by a vertically tensioned cable) to approximate lunar gravity, a continuous excavator maintains steady production while a discrete scraper's mobility degrades precipitously.

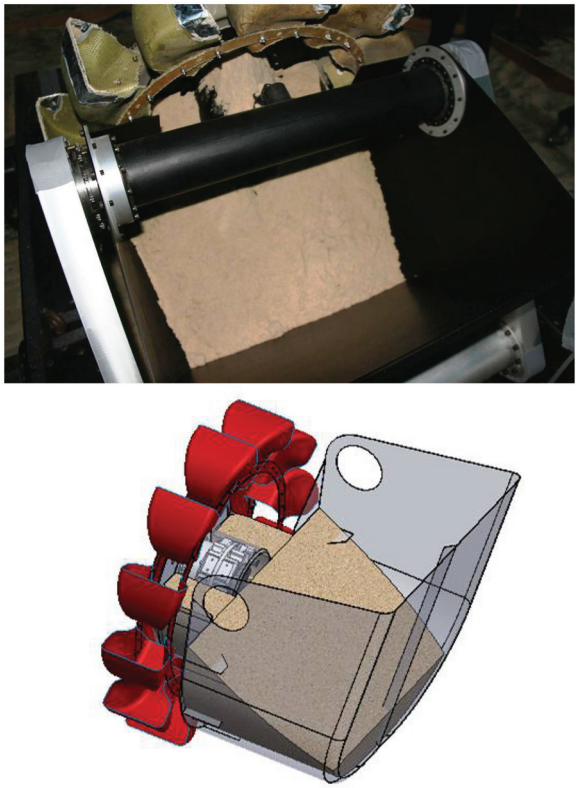

Fig. 16. Sand in dump-bed, having been transferred directly by gravity from bucket-wheel (top); also shown in CAD (bottom).

The analysis that indicates higher values of L for continuous excavation than for discrete scraping can be reduced to three characteristics:

1. the excavation resistance coefficient as a function of payload ratio, $\tilde{F}(\tilde{P})$, can be approximated linearly for discrete scraping;
2. $\tilde{F}(\tilde{P})$ can be approximated by a zero-slope line for continuous excavation;
3. the non-dimensional slope of $\tilde{F}(\tilde{P})$ for discrete scraping (i.e. $\tilde{F}'_{disc}$) is more than three times larger than the non-dimensional y-intercept of $\tilde{F}(\tilde{P})$ for continuous excavation (i.e. $\tilde{F}_{0,cont}$).

All three of these characteristics are demonstrated experimentally in this work, and are shown to be insensitive to scale. The experimental results demonstrating characteristic (1) are consistent with experiments by other researchers (Agui and Wilkinson, 2010). Characteristic (2) agrees with the logic that follows from empty buckets regularly providing resets of the excavation resistance encountered. Characteristic (3) is demonstrated by experiments with various continuous and discrete excavation tools. Gravity-offloaded robotic excavation experiments directly show that a particular discrete excavator configuration encounters $L < 1$ while a continuous excavator configuration does not, implying that $\tilde{F}'_{disc}$ is larger than $\tilde{F}_{0,cont}$ by a factor of at least three; the factor is estimated to be approximately 30. Independent benchtop experiments with flat-plates and simple bucket-wheels consistently show $\tilde{F}'_{disc}$ to be larger than $\tilde{F}_{0,cont}$ by a factor estimated to be between 40 and 60, at multiple scales. Although these experimental results do not outright prove characteristic (3) generally, the consistency between entirely different experiments and the magnitude of the margin by which $\tilde{F}'_{disc}$ is regularly larger than $\tilde{F}_{0,cont}$ strongly suggest that the characteristic is likely to hold in all but perhaps the most degenerate of cases.

Novel experiments that offload the weight of excavators (by applying an offset force of five-sixths of the vehicle weight with a vertical cable) while they dig show the risks of estimating low-gravity performance based purely on nominal testing in 1 g. Tests in 1 g show both discrete and continuous excavators achieving productive excavation, but in offloaded tests the same discrete excavator gets bogged down and fails to collect a significant amount of regolith payload. Although not as accurate a representation of low-gravity environments as reduced-gravity flights, where the gravitational acceleration acting on the soil is also offset, offloaded tests provide a more balanced estimate of planetary excavation than nominal testing in 1 g. Offloaded testing underestimates the detrimental effects of low gravity on traction but overestimates the detrimental effects of low gravity on excavation resistance. Testing in 1 g, on the other hand, underestimates the detrimental effects of low gravity both on traction and on excavation resistance. Thus, 1 g testing is more likely to underestimate the overall challenges of low-gravity excavation than offloaded testing. As a result, offloaded excavation testing provides valuable additional information when evaluating planetary excavators envisioned for low-gravity operations.

Experiments were conducted in GRC-1, a washed industrial silica-sand devoid of agglutinates and of the sub-75 micron basaltic fines that make up 40% of lunar regolith. The important dangers related to dust generation and dust contamination tolerance are therefore not directly addressed. The achieved densities for experimentation are 1640 kg/m^3 (very loose/loose) and 1720 kg/m^3 (medium dense).

A novel excavator configuration with direct transfer from a transverse bucket-wheel into a collection bin/dump-bed addresses practical challenges of developing continuous excavators for planetary environments. The bucket-wheel is a single moving part, with no need for chains or conveyors that introduce additional mechanical failure risks when exposed to a harsh vacuum, regolith and dust. This configuration offers a solution to the transfer problem for bucket-wheels identified in past literature.

Directions for future work include better understanding low-gravity excavation and mobility by means of experiments where the excavator and regolith are both subject to reduced gravity, such as reduced-gravity flights or controlled drop towers. Another important direction of future work is comparing continuous excavation to discrete *scooping* (which includes more general bucket motions).

Such an approach could utilize a higher fraction of the robot weight by incorporating a vertical penetration component in the bucket motion followed by actuated curling to break out of the regolith; this could potentially overcome the limitations of discrete *scraping*, which has been the class of discrete excavation considered for planetary excavation to date. It is important to note that percussive excavation is particularly effective for discrete scooping approaches that include significant penetration components.

The study of excavation in the presence of rocks is suggested as another future direction of work. Full-scale excavation task experimentation (full load-haul-dump cycles) with gravity offload is also suggested for future work, as well as work on task autonomy (perhaps supervised autonomy with time delay).

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